

Quiz 2

Morning

$$Q \text{ No 1: } P(\text{incorrect diagnosis}) = 1 - 0.7 = 0.3 = P(B)$$

$$P(\text{patient sues} | \text{incorrect diagnosis}) = 0.9 = P(A|B)$$

$$P(\text{incorrect diagnosis} \cap \text{patient sues}) = ? = P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B) \times P(A|B) = P(A \cap B)$$

$$0.3 \times 0.9 = P(A \cap B)$$

$$0.27 = P(A \cap B)$$

$$Q \text{ No 2: } P(\text{Departs on time}) = 1 - 0.17 = 0.83 = P(D)$$

$$P(\text{Arrives on time}) = 1 - 0.18 = 0.82 = P(A)$$

$$P(\text{Departs and arrives on time}) = 0.78 = P(D \cap A)$$

$$P(A|D) = \frac{P(D \cap A)}{P(D)} = \frac{0.78}{0.83} = 0.94$$

Q No 3:

$$Q \quad C = \left[A^{(1:2:, :)} ; B^{(2:2:, :)} \right]$$

Q No 4:

$$n = \text{fix}(20 \times \text{rand}(15, 1))$$

Quiz 2

Afternoon

Q no 1 : $P(\text{Married Man watches show}) = 0.4 = P(B)$

$$P(\text{married women watches show}) = 0.5 = P(A)$$

$$P(\text{man watches} \mid \text{women watches}) = 0.7 = P(B|A)$$

$$P(\text{wife watches} \mid \text{husband watches}) = ? = P(A|B)$$

To find $P(A \cap B)$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \Rightarrow P(B|A) \times P(A) = P(A \cap B)$$

$$P(A \cap B) = 0.7 \times 0.5 = 0.35$$

To find $P(A|B)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.35}{0.4} = 0.875$$

Q no 2: $P(\text{Departs on time}) = 1 - 0.17 = 0.83 = P(D)$

$$P(\text{Arrives on time}) = 1 - 0.18 = 0.82 = P(A)$$

$$P(\text{Departs on time} \cap \text{arrives on time}) = 0.78 = P(D \cap A)$$

$$P(D|A) = \frac{P(D \cap A)}{P(A)} = \frac{0.78}{0.82} = 0.95$$

Q no 3: $C = [A(:, 2:2:) \ B(:, 1:2:)]$

Q no 4: $n = \text{fix}(30 \times \text{rand}(1, 10))$